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CS 360: Machine Learning Lab
Evaluation Assignment 2

Instructions: This assignment is for evaluation, and marks will be awarded. You need to complete execution by 12 PM. The saved results folder (consisting of .csv files, plots, and a Word file) has to be submitted through a Google form, which will be shared by the teaching assistant.

Dataset requirements: You may use any dataset that is suitable for K-Means clustering (for example, from the UC Irvine Machine Learning Repository, Kaggle, or scikit-learn). The dataset must satisfy all of the following:

- All features used for clustering are numerical, with at least 200 samples and at least 5 features
- The dataset is reasonably suited to K-Means (roughly compact, non-overlapping-by-design groups). Justify briefly in your report why it is suitable.

Note: Datasets already used in previous lab assignments are strongly discouraged. Marks may be deducted if a previously used dataset is chosen. State the dataset name, number of samples, and number of features at the start of your report.

1. Write a program to perform the following tasks:

- Apply K-Means for $K = 2, 3, \dots, 10$ on the dataset, and record the SSD and the average silhouette score for each K in a table.
- Plot SSD vs. K (Elbow method) and average silhouette score vs. K . Use both to select a suitable value of K . If the two criteria disagree, explain how you resolved the disagreement (e.g., using per-cluster silhouette plots, cluster sizes, or domain knowledge). Justify your final choice.
- Visualize the clusters for the chosen K . If the data has more than two dimensions, randomly select any two features, colour points by cluster, and mark the centroids. Report the percentage of variance explained by each of the two components.
- For the final clustering (chosen K), report in a table:
 - the final cluster centroids
 - the size of each cluster
 - the SSD, total and per cluster
 - the average silhouette score (and the per-cluster average silhouette).
- Repeat parts (b) and (c) on the scaled data using the same settings. Compare the elbow plots, silhouette curves, selected K , and final cluster sizes against the scaled version. Present the comparison in one table or side-by-side plots.
- Write an analysis:
 - The quality of the obtained clusters (cohesion, separation, balance of sizes);
 - The sensitivity of the results to the choice of K and to the random initialization (e.g., re-run with 5 different `random_state` values and report the range of SSD and silhouette);
 - The observed effect of feature scaling, referring to your experiment in part (e);